

Vikas Dwivedi

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Career Summary

I hold a PhD from IIT Madras and a Master's from IIT Delhi. Professionally, I bring over two years of industry experience in Data Science and nearly three years of postdoctoral research in India, France, and the US. My research focuses on domain-informed AI, combining data, machine learning, and domain knowledge, with interests in computational fluid dynamics, scientific machine learning, and computer vision for biomedical and atmospheric applications.

PhD Details

Degree	Institute	Supervisor	Thesis Title	Duration
PhD (Mechanical)	Indian Institute of Technology-Madras	Prof Balaji Srinivasan (Deptt of Mechanical Engg and Wadhvani School of Data Science and AI)	Physics Enhanced Machine Learning Methods for Flow Simulation (<i><u>Best PhD Thesis Award in Data Science-2021</u></i>)	Jan 15-Mar 21

Educational Details

S.No.	Degree/Exam	Institute/Board	Grade	Duration
1	M.Tech (Applied Mechanics)	Indian Institute of Technology-Delhi	8.4	Jul 12-Nov 14
2	B.E. (Mechanical)	Institute of Engineering and Technology-DAVV, Indore	70%	Jul 07-May 11
3	10 th / 12 th	Central Board of Secondary Education	84.4% / 80.2%	2004 /2006

Postdoctoral Experience

S.No.	Position	Institute	Research Topic	Duration
1	Postdoctoral Researcher	Creatis Biomedical Imaging Laboratory, INSA-Lyon, France	Biophysics-informed medical imaging	Nov 24-Now (18 months)
2	Postdoctoral Researcher	State University of New York at Albany, USA	Machine Learning (ML)-based tropical cyclone intensity estimation	Jul 23-May 24 (11 months)
3	Postdoctoral Researcher	Robert Bosch Center of Data Science and AI at IIT-Madras, India	Physics-informed medical image segmentation	Jul 22-Mar 23 (9 months)

Industry Experience

S.No.	Position	Institute/Industry	Research Topic	Duration
1	Data Scientist	Vunet Systems, Bangalore	ML analysis of business/performance observability platforms	Apr 20-Jun 22 (27 months)

Publications in Indexed Journals

S.No.	Paper Link	Year	Authors	Journal	Citations
1	Meta-SPINN: Meta-learned basis adaptation for parametric PDEs [Link]	2026	Vikas Dwivedi , Monica Sigovan and Bruno Sixou	<i>Machine Learning: Science and Technology (IOP)</i> IF=4.2, Q2	0
2	Gated X-TFC: Soft domain decomposition for forward and inverse problems in sharp-gradient PDEs [Link]	2026	Vikas Dwivedi , Enrico Schiassi, Bruno Sixou and Monica Sigovan	<i>Neurocomputing (Elsevier)</i> , IF=6.5, Q1	3
3	Curriculum learning-Driven PIELMs for fluid flow simulations [Link]	2025	Vikas Dwivedi , Bruno Sixou and Monica Sigovan	<i>Neurocomputing (Elsevier)</i> , IF=6.5, Q1	10
4	Physics informed contour selection for rapid image segmentation [Link]	2024	Vikas Dwivedi , Balaji Srinivasan and Ganapathy Krishnamurthi	Scientific Reports (<i>Nature</i>), IF=3.9, Q1	10

5	A normal equation-based extreme learning machine for solving linear partial differential equations [Link]	2021	Vikas Dwivedi and Balaji Srinivasan	Journal of Computing and Info Science in Engineering (<i>American Society of Mech Engineers</i>), IF=3.3, Q1	30
6	Distributed learning machines for solving forward and inverse problems in partial differential equations [Link]	2020	Vikas Dwivedi , Nishant Parashar and Balaji Srinivasan	Neurocomputing (<i>Elsevier</i>), IF=6.5, Q1	137
7	Solution of biharmonic equation in complicated geometries with physics informed extreme learning machine [Link]	2020	Vikas Dwivedi and Balaji Srinivasan	Journal of Computing and Info Science in Engineering (<i>American Society of Mech Engineers</i>), IF=3.3, Q1	35
8	Physics informed extreme learning machine (pielm)–a rapid method for the numerical solution of partial differential equations [Link]	2019	Vikas Dwivedi and Balaji Srinivasan	Neurocomputing (<i>Elsevier</i>), IF=6.5, Q1	331
9	Prediction of transition in temporal mixing layer using ILES [Link]	2018	Vikas Dwivedi and Balaji Srinivasan	International Journal of Advances in Engineering Sciences and Applied Mathematics (<i>Springer</i>), IF=1.1	1

Publications in Indexed Conference Proceedings

S.No.	Title	Year	Authors	Conference	Proceedings
1	Towards Sustainable Scientific Machine Learning: Fast and Interpretable PDE Solvers via RBF-PIELM [Link]	17-20 Dec 2025	Akshay Govind Srinivasan, Vikas Dwivedi and Balaji Srinivasan	13 th ACM IKDD International Conference on Data Science	Association of Computing Machinery
2	Global versus Local: Evaluating AlexNet Architectures for Tropical Cyclone Intensity Estimation [Link]	01-05 Dec 2024	Vikas Dwivedi	27 th International Conference of Pattern Recognition	Lecture Notes in Computer Science (<i>Springer</i>)

Presentations in Reputed International Conferences

S.No.	Title	Year	Authors	Conference
1	Kernel-Adaptive Physics-Informed Shallow Meta-Learning for Parametric Linear PDEs [Link]	26-27 Apr 2026	Vikas Dwivedi , Bruno Sixou and Monica Sigovan	AI & PDE: ICLR 2026 Workshop on AI and Partial Differential Equations, Rio de Janeiro, Brazil
2	AB-PIELMS: Adaptive-Basis Physics-Informed Extreme Learning Machines for residual-driven domain decomposition [Link]	26-27 Apr 2026	Aldis Daniel, Vikas Dwivedi , Balaji Srinivasan	AI & PDE: ICLR 2026 Workshop on AI and Partial Differential Equations, Rio de Janeiro, Brazil
3	Curriculum-Learning PIELMs for Hemodynamic Flows [Link]	6 Dec 2025	Vikas Dwivedi , Bruno Sixou and Monica Sigovan	17 th Annual Workshop on Optimization for Machine Learning (NeurIPS-OPT 2025), San Diego, USA
4	Empirical-Bayes XTFC for Inverse Parameter Estimation [Link]	6 Dec 2025	Vikas Dwivedi , Bruno Sixou and Monica Sigovan	17 th Annual Workshop on Optimization for Machine Learning (NeurIPS-OPT 2025), San Diego, USA
5	Deep vs. Shallow: Benchmarking Physics-Informed Neural Architectures on the Biharmonic Equation [Link]	6 Dec 2025	Akshay Govind Srinivasan, Vikas Dwivedi and Balaji Srinivasan	8 th Annual Workshop on Machine Learning and Physical Sciences (NeurIPS-ML4PS 2025), San Diego, USA
6	Numerical experiments to solve partial differential equations with a denoising auto-encoder	09-11 Dec 2020	Vikas Dwivedi and Balaji Srinivasan	8 th International conference on fluid mechanics and fluid power, IIT-Guwahati, India
7	Rapid solution of PDEs in complex geometries with physics informed extreme learning machine	02-04 Dec 2019	Vikas Dwivedi and Balaji Srinivasan	1 st Indo-German conference on computational mathematics, IISc-Bangalore, India
8	Data driven solution of linear diffusion equation with distributed physics informed neural networks	03-04 Sept 2019	Vikas Dwivedi and Balaji Srinivasan	7 th Asian symposium of computational heat and mass transfer, Tokyo University of Science, Japan

Journal Articles (Revision Under Review)

S.No.	Title	Year	Authors	Journal
1	Kernel-Adaptive PI-ELMs for Forward and Inverse Problems in PDEs with Sharp Gradients [Link]	2025	Vikas Dwivedi , Balaji Srinivasan, Bruno Sixou and Monica Sigovan	<i>Neural Networks (Elsevier)</i>

Other Information

Category	Description
Award	Best PhD Thesis Award in Data Science-2021 by IIT Madras
Invited Lecture	Scientific Machine Learning for Fluid Mechanics at IIT Kanpur on 6 Nov 23
Keynote Speaker	International Conference in Applied Mechanics, Machine Learning and Applied Computation-2022 at NIT Raipur
MOOC	Taught a short term course on Data-Centric AI at NPTEL
Reviewer	Neurocomputing, Scientific Reports, Applied Mathematical Modeling
Test Score (GATE)	All India Rank 325 among 81175 students in GATE-2011 (Mechanical Engineering)
Scholarship	Received MHRD scholarship during masters and PhD

Referees

S.No.	Referee	Designation	Email
1	Prof Balaji Srinivasan	Professor, Department of Mechanical Engineering and Wadhwani School of Data Science and AI, Indian Institute of Technology, Madras	sbalaji@dsai.iitm.ac.in
2	Prof Ganapathy Krishnamurthi	Professor, Department of Engineering Design and Wadhwani School of Data Science and AI, Indian Institute of Technology, Madras	gankrish@dsai.iitm.ac.in
3	Prof Bruno Sixou	Associate Professor, Creatis Biomedical Imaging Laboratory, Institut National des Sciences Appliquées de Lyon (INSA-Lyon), France	bruno.sixou@insa-lyon.fr
4	Dr Monica Sigovan	CNRS Research Scientist, Creatis Biomedical Imaging Laboratory, Institut National des Sciences Appliquées de Lyon (INSA-Lyon), France	monica.sigovan@insa-lyon.fr
5	Mr Jithesh Kaveetil	Co-Founder and CTO, Vunet Systems, Bangalore	jithesh@vunetsystems.com